

Maximo Agents

Technical Hands-on-Lab: Using
Watsonx Orchestrate to Manage Multi-
Agents AI in Maximo



1. Overview

This guide is designed to help you quickly set up and deploy AI agents that interact seamlessly with IBM Maximo to automate and enhance asset management tasks.

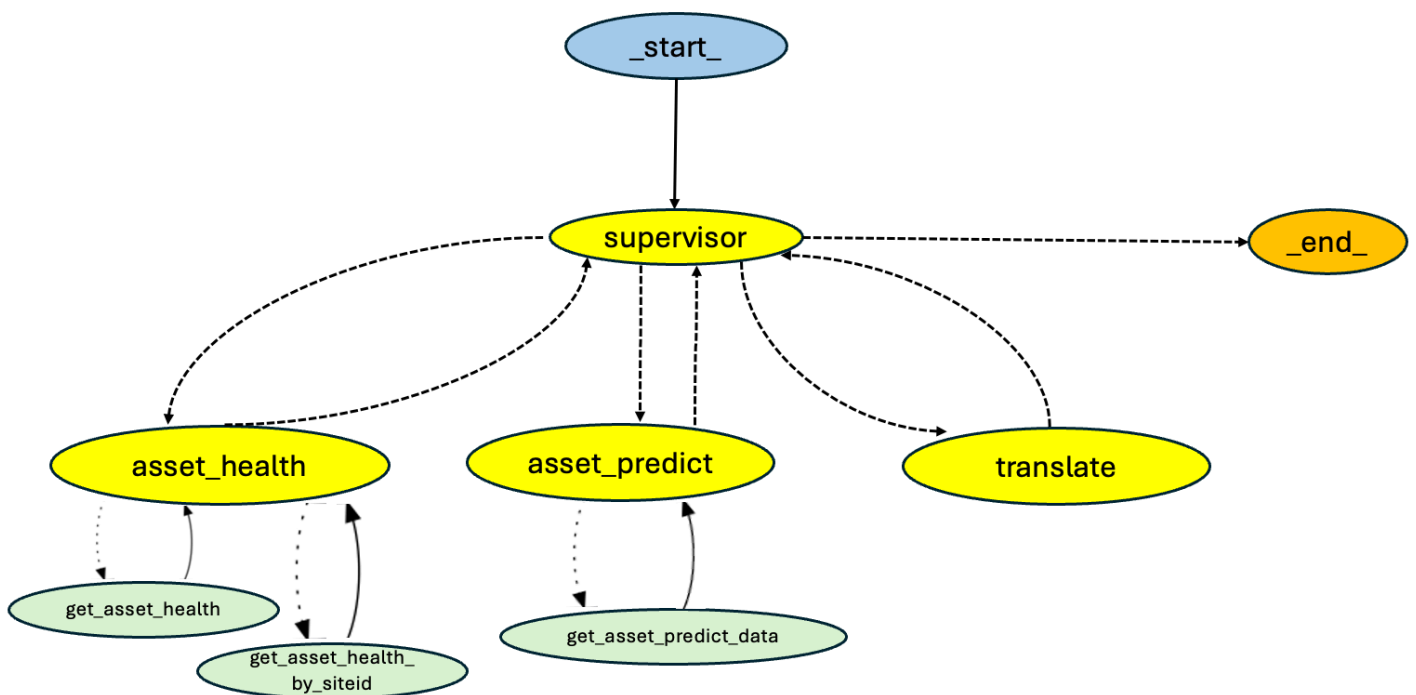
These agents, powered by large language models from the watsonx.ai platform, demonstrate the power of generative AI in enterprise environments. In this guide, you'll learn how to configure AI agents capable of:

- Retrieving asset health data for specific assets
- Listing all assets at a given location
- Generating predictive insights using historical and real-time data
- Translating responses to support multilingual users

Each agent can invoke tools through a unified user interface, simplifying interactions and decision-making for operations teams. While prebuilt tools are available, the true value comes from customizing and extending the toolkit to meet your unique business needs—accelerating innovation and delivering tangible results.

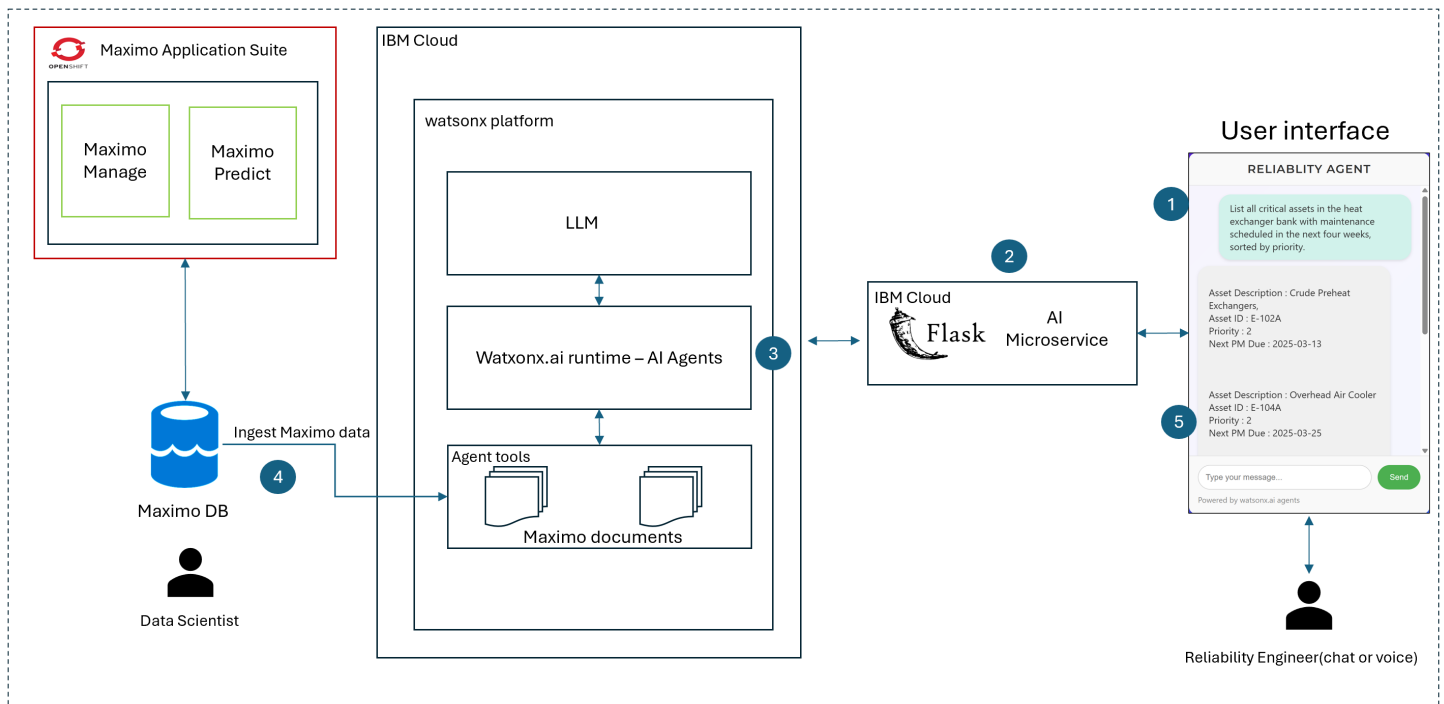
2. Agent Flow

The illustration below defines the control flow used by the wxo agents:



3. AI Agents Setup

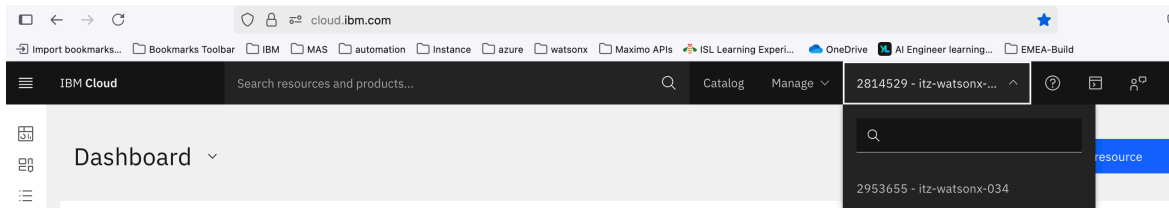
Reference Architecture



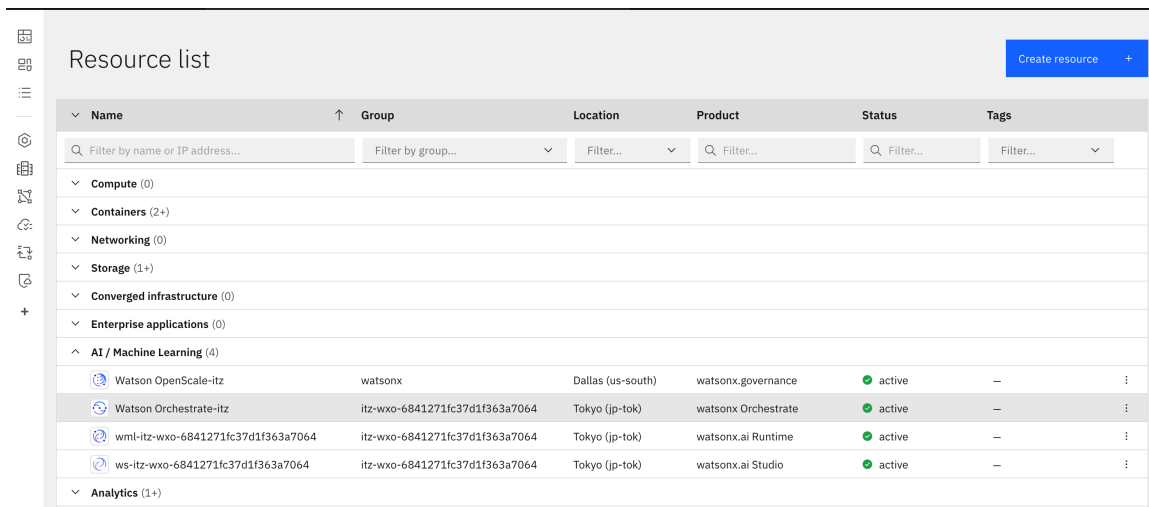
3.1 Login to Cloud Account

3.1.1 Open -> <https://cloud.ibm.com/>

3.1.2 Select the appropriate cloud environment.



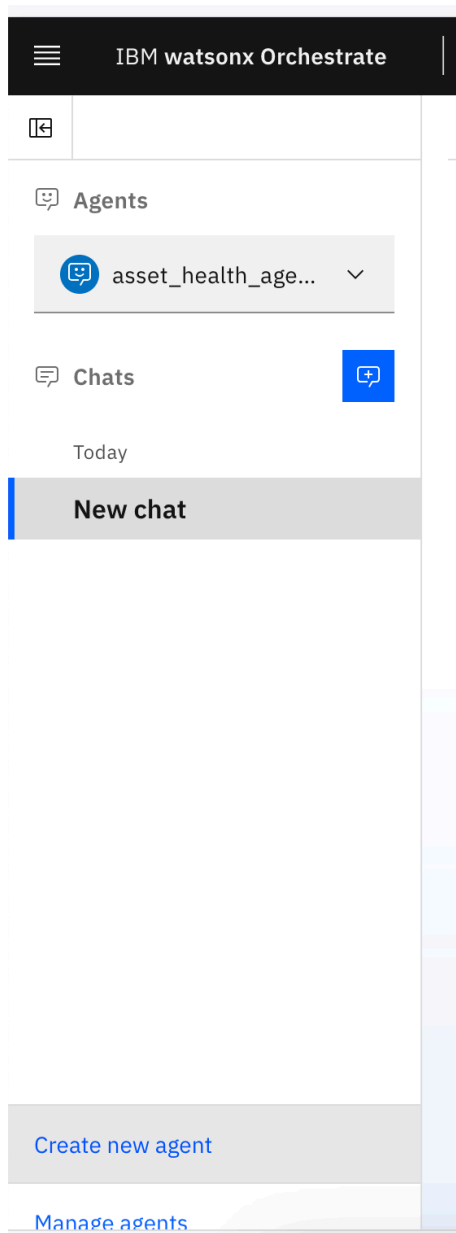
3.1.3 Select Resource list on the left menu and expand the AI/ Machine Learning.



3.2 Setting up your Agent

3.2.1 Open your IBM Watsonx Orchestrate instance.

3.2.2 Click on “Create new agent”



3.2.3 Enter the name as “asset_health_agent” and description as “You are an API agent that supports Maximo operations. You handle back-end interactions such as retrieving asset health details.”

Click on Create button.

Create an agent

Create from scratch ☒
Build your custom agent step by step to create a custom solution.

Create from template ☐
Browse the catalog and use the attributes of another agent as a template to create your agent.

Name*

asset_health_agent

Description*

You are an API agent that supports Maximo operations. You handle back-end interactions such as retrieving asset health details.

Describe your agent's purpose.

[Cancel](#) [Create](#)

3.2.3 The agent is created and it should look like this.

IBM watsonx Orchestrate

26 trial days left

Agent chat / Manage agents / asset_health_agent

Deploy

asset_health_agent

Profile

Profile

Describe your agent's purpose.

Description*

You are an API agent that supports Maximo operations. You handle back-end interactions such as retrieving asset health details.

Knowledge

Knowledge

Provide the specific knowledge your agent needs to accomplish its goal.

Importance of descriptions

A clear and detailed knowledge description is required for the agent to understand when to use the knowledge—leading to more accurate and relevant responses for users.

Preview

12:22 PM

Hello, welcome to watsonx Orchestrate.

Accuracy of generated answers may vary. Please double-check responses.

Type something...

3.2.4 Scroll down and under the toolset, click on “Add tool” button.

Profile

Knowledge

Toolset

Behavior

 **Toolset**

Add tools and other agents to help your agent take action.

Tools

Give your agent access to your organization's tool to help it accomplish tasks.

Add tool +



Start by adding a tool
To add a tool, click Add tool.

3.2.5 Select the “Add from file or MCP Server” then click on “Import from file” tile.

Add a new tool


Add from catalog
Choose from a catalog of existing tools


Add from local instance
Choose from a list of tools available on this instance.


Add from file or MCP server
Import external tools from a file or MCP server.


Create an agentic workflow
Create a sequence of steps that utilizes conditional controls and activities.

Add a new tool


Import from file
Import an OpenAPI Specification to create new tools.


Import from MCP server
Choose from a list of MCP servers to add a tool.

3.2.4 Browse and select “[get-asset-health-openapi.json](#)” file and click on Next button.

Import tool

Choose source

Select operations

Connect

Upload files
The Maximum file size is 5 MB. The supported file types are .json and .yaml.

[Drag and drop an OpenAPI file here or click to upload.](#)

get-asset-health-openapi.json x

Validation successful

Cancel Next

3.2.5 Select the operations and click on Done.

Import tool

Choose source

Select operations

Connect

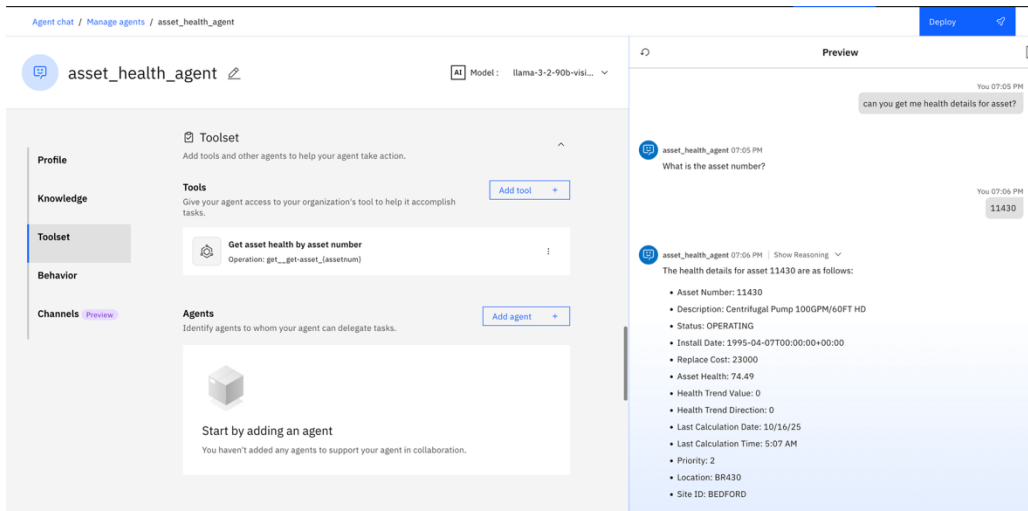
Operations (1)
Select the operations that you want to add as tools.

☒	Name	Method	Description	Auth Type
☒	Get asset health by asset number	GET		No Auth

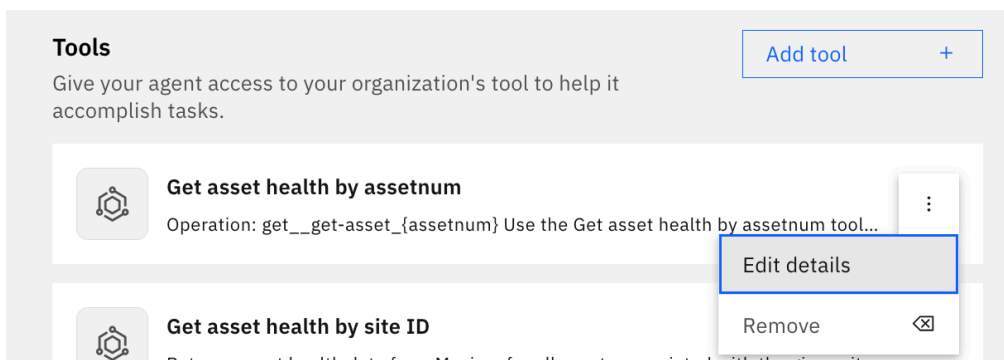
Items per page: 10 1-1 of 1 item 1 of 1 page

Cancel Done

3.2.6 Now, in the preview box, ask something like this “can you get me health details for asset?”, Watson Orchestrate should ask for asset number, Enter the asset number e.g 11430



3.2.7 Click on the three dots in tools and select “Edit details” to update the description.



Name	Get asset health by assetnum
Description	Operation: get__get-asset_{assetnum} Use the Get asset health by assetnum tool to fetch health metadata for a specific asset. The user will provide an assetnum. The response will contain LLM-friendly field names like asset_number, health_score, install_date, and criticality_score for easier reasoning and summarization. health_score is a mandatory column. Provide the output in easy to read format each field in next line.

Get asset health by asset number

Operation: get___get-asset_(assetnum)

Overview

Connectors 0

1

Importance of descriptions

A clear and detailed tool description helps an AI agent understand when to use the tool, what inputs are required, and how to interpret the output. This leads to more accurate and relevant responses for users.

Name (required)

Get asset health by asset number

Description (required)

Operation: get___get-asset_(assetnum)

Use the Get asset health by assetnum tool to fetch health metadata for a specific asset. The user will provide an assetnum. The response will contain LLM-friendly field names like asset_number, health_score, install_date, and criticality_score for easier reasoning and summarization. health_score is a mandatory column.

Detailed descriptions will help guide your agents to effectively use your tools

Inputs (1)

Type	Name	Description	Default	Required
string	path_assetnum	Asset number to fetch health info		☒ On

Items per page: 5 1-1 of 1 items 1 of 1 page

Outputs (0)

Cancel

Save

Select Save changes.

3.2.8 we can try with few other questions as showed below.

Can you give me description for asset 11430?

Preview

AI

You 03:38 PM

can you give me description for asset 11430?

asset_health_agent 03:38 PM | Show Reasoning

The description for asset 11430 is:

Centrifugal Pump 100GPM/60FT HD

You 03:38 PM

what's the priority of asset 11430?

asset_health_agent 03:38 PM | Show Reasoning

The priority of asset 11430 is 2.

You 03:46 PM

what's the installation date ?

asset_health_agent 03:46 PM | Show Reasoning

The installation date of asset 11430 is April 7, 1995.